Business Requirements Document (BRD)

Example Autonomous Warehouse Inspection System

Document Control	Value
Document ID	BRD-AWIS-001
Version	1.0
Status	Example / Training Reference
Owner	Business Sponsor / Product Owner
Purpose	Define WHY the initiative is needed and WHAT business/operational outcomes the solution must achieve.

1. Executive Summary

ABC Logistics operates three distribution centers. Current warehouse inspections depend on manual patrols, spreadsheets, and delayed escalation. The proposed initiative will introduce an autonomous inspection capability to improve inspection frequency, inventory visibility, safety, and operating economics.

2. Business Context and Problem

- Manual inspections consume approximately 160 labor-hours per warehouse per week.
- Inspection quality varies by shift and operator; some aisles are inspected less frequently during peak periods.
- Blocked aisles, damaged pallets, and misplaced inventory may remain undetected for several hours.
- Management lacks a consistent digital audit trail linking inspection findings, corrective actions, and closure.

3. Scope

In Scope	Out of Scope
Scheduled autonomous aisle inspection	Automated picking and packing
Detection/reporting of defined abnormalities	Forklift replacement
Operator alerts and management dashboard	Replacement of ERP/WMS
Inspection history and KPI reporting	Outdoor yard operations

4. Commercial / Economic Business Requirements

These requirements establish the economic justification and commercial boundaries of the initiative. Each requirement is measurable and verified against an agreed baseline.

ID	Requirement	Measure / Acceptance Target
BR-C01	Reduce direct labor used for routine warehouse inspection.	At least 40% reduction versus approved 2026 baseline after 3 months stable operation.
BR-C02	Achieve acceptable financial payback.	Simple payback period <=24 months using approved CAPEX/OPEX assumptions.
BR-C03	Limit recurring operating cost.	Annual software, support, maintenance and consumables <=15% of initial acquisition cost.
BR-C04	Reduce inventory-related operational loss.	At least 25% reduction in losses attributable to late detection within 12 months.
BR-C05	Enable growth without proportional staffing.	Doubling inspected floor area requires <=20% increase in inspection-related staffing.
BR-C06	Support phased deployment.	Pilot assets reusable in production; <=10% pilot hardware write-off.

5. Business Value Measurement

KPI	Baseline	Target	Source
Inspection labor	160 hr/week/site	<=96 hr/week/site	Timekeeping
Inspection cost	₩180M/site/year	<=₩108M	Finance
Issue discovery delay	120 min	<=30 min	Incident records
Coverage	70% / 4 hr	100% / 4 hr	Mission records

6. Technical / Operational Business Requirements

These define WHAT the business expects the solution to accomplish and the measurable service outcome. They do not prescribe robot, sensor, AI model, middleware, or database selections.

ID	Operational Requirement	Measurable Goal
BR-T01	Inspect all designated warehouse aisles.	100% of active inspection zones at least once every 4 hours.
BR-T02	Identify designated damaged-pallet conditions.	Precision >=95% and recall >=90% on approved operational validation set.
BR-T03	Identify blocked travel aisles.	Detect defined aisle-blocking conditions with >=98% detection rate.
BR-T04	Identify defined misplaced-inventory conditions.	>=95% correct identification in approved acceptance scenarios.
BR-T05	Escalate critical safety findings.	Responsible operator notified <=5 seconds after confirmed finding.
BR-T06	Complete scheduled missions without routine human intervention.	>=98% mission completion over a 30-day production-equivalent trial.
BR-T07	Provide inspection history.	100% completed missions/findings searchable for >=12 months.
BR-T08	Remain available during scheduled periods.	>=99.5% service availability excluding approved maintenance.

7. Stakeholders and User Groups

Stakeholder	Need / Decision Interest
Executive Sponsor	ROI, risk, scalability
Warehouse Manager	Coverage, productivity
Safety Manager	Safe behavior, auditability
Operators	Clear alerts and recovery
IT / OT Support	Supportability, cybersecurity
Finance	CAPEX/OPEX and benefit realization

8. Business Rules and Operational Constraints

ID	Rule / Constraint
BR-R01	Critical safety findings remain open until acknowledged by an authorized supervisor.
BR-R02	Autonomous inspection shall not obstruct emergency routes or materially delay warehouse traffic.
BR-R03	Only authorized roles may modify inspection zones, schedules, or finding severity.
BR-R04	Evidence used for KPI calculation shall be retained and auditable.
BR-R05	Deployment shall comply with site safety policies and applicable requirements.
BR-R06	The business shall define the approved defect taxonomy and validation dataset before final validation.

9. Assumptions, Dependencies and Risks

Type	Example
Assumption	Floor, aisle markings and lighting remain within agreed conditions.
Dependency	Current warehouse layout and inventory master data are maintained.
Dependency	Wireless and charging infrastructure may be upgraded if SRD requires.
Risk	Frequent layout changes may reduce mission success.
Risk	Poor defect labels may prevent objective AI acceptance testing.
Risk	Labor savings may not be realized if operating procedures are not redesigned.

10. Business Acceptance

Acceptance occurs when all High-priority commercial/economic and technical/operational requirements meet their stated measures during the agreed pilot or production-equivalent evaluation and no unresolved critical safety/compliance issue remains.

11. Example Use Cases

Use Case	Business-Level Description
UC-01 Scheduled Inspection	Supervisor defines approved zones/schedule; solution completes inspection and records results without routine intervention.
UC-02 Critical Finding	Solution records evidence, alerts the responsible role within target time, and tracks acknowledgement.
UC-03 Management Review	Manager reviews coverage, open findings, trends and economic KPIs.
UC-04 Recovery / Exception	Incomplete inspection is recorded and controlled rescheduling/operator intervention is enabled.

12. BRD-to-SRD Traceability Examples

Business Requirement	What the SRD must determine
BR-T01: 100% zones / 4 hr	Robot quantity/type, speed, runtime, charging and fleet capacity.
BR-T02: >=95% precision / >=90% recall	Sensor, AI model, compute throughput and image quality.
BR-T05: alert <=5 sec	Network latency/capacity, server and notification platform.
BR-T08: availability >=99.5%	Reserve capacity, maintainability, monitoring and charging.
BR-C02: payback <=24 months	Technology selections must remain within derived cost envelope.

13. Glossary

BRD: Business Requirements Document. KPI: Key Performance Indicator. CAPEX: Capital Expenditure. OPEX: Operating Expenditure. Precision/Recall: measured using the approved validation dataset.